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RIPARIAN LAND-COVER CLASSIFICATION METHODOLOGY OF AIRBORNE LASER BATHYMETRY -ASSISTED DEEP LEARNING OF AERIAL PHOTOGRAPHS

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Abstract

Land cover classification (LCC) has been acting as a fundamental character in hydraulic research works and practical tasks in terms of both flood control and riverside environment conservation. For this work, we developed a new methodology for mapping the LCC (water area, bare ground, herbaceous species, arborous species, and bamboo grove) in riparian area. In this work, we upgraded the technique for incorporating the ALB dataset with an auxiliary module, which was interpolated in the decoder module of the Deeplabv3+ model, comparing with the existing analysis results based on the ALB dataset only (the same data as our method) for the LCC mapping, all the recall values of our methodology for the vegetation classification have been improved approximately 20%.

Keywords: Riparian land-cover classification, ALB, deep learning, semantic segmentation

Introduction

Recently, the detailed data for the river cross-section and those attributes have been measured and estimated using effective remote sensing methods (RSM) like airborne laser bathymetry (ALB), some hydraulic researchers have attempted to evaluate LCC of the Asahi river riparian area using the traditional analysis method (ALB dataset only). In this work, we have modified the deeplabv3+ module that can use the ALB dataset to interpolate. We applied our methodology to ALB dataset measured in March, July, November 2017 for the approximately 4.5 km-long downstream area of the Asahi River, Japan.

Methods

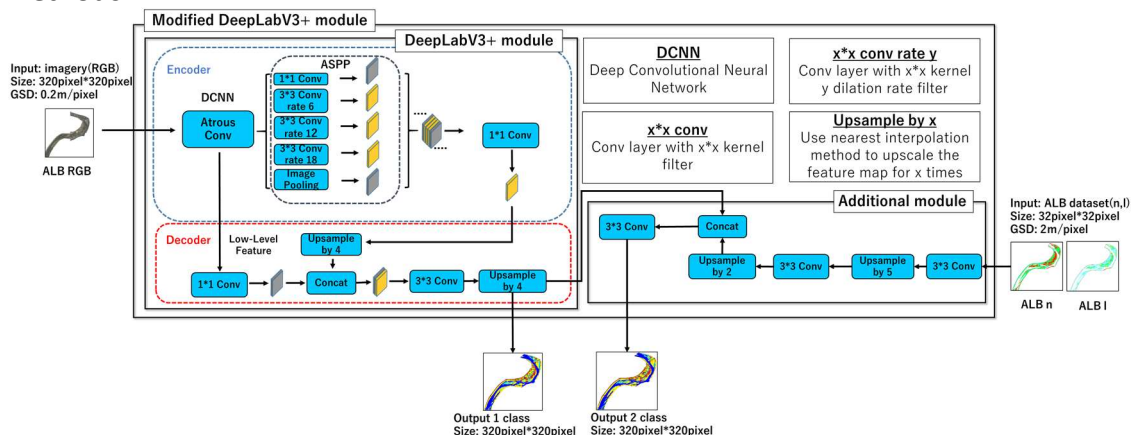


Fig. 1. Deep learning process of imagery with ALB dataset (DCNN part with Xception10 backbone)

For this work, the image processing technique of Deeplabv3+ (Chen *et al.*, 2018) for semantic segmentation was used for aerial photographs taken during the ALB measurement period. Additionally, we combined the ALB dataset: the voxeled laser point cloud (n) and vegetation

height (I) (digital surface model data minus digital terrain model data) in 2m*2m mesh. Appending an auxiliary module to the original technique, we concatenated these two types of data (aerial photographs and ALB dataset). Herein, we set the resolution ratio of 320pixel*320pixel (GSD:0.2m/pixel) for aerial photographs and 32pixel*32pixel (GSD:2m/pixel) for ALB dataset as 1:10, which can interpolate the ALB dataset in the aerial photographs as the same size.

Results

Results revealed that our method predicts the LCC map more precisely in case of use of combined dataset, especially for the detection of herbaceous species, comparing with predictions using aerial photographs only. Furthermore, comparing with existing analysis results based on the ALB dataset only (the same data as our method) for the LCC mapping, all the recall values of our methodology for the vegetation classification have been improved over 20% (herbaceous species for 25.2%, arborous species for 20.5%, and bamboo grove for 22.7%).

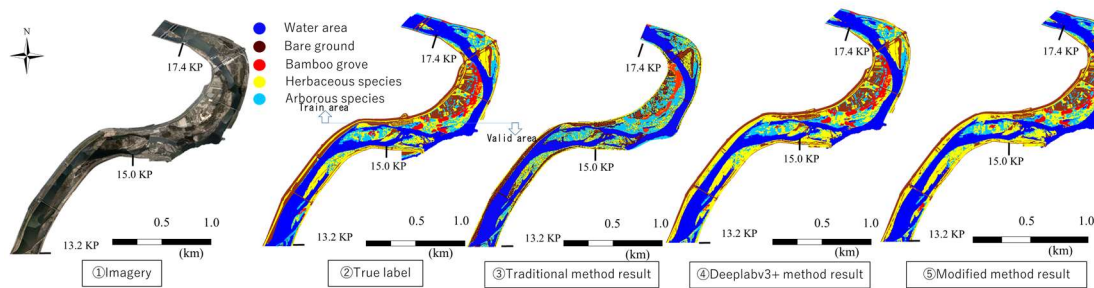


Fig. 2. Comparison of each method result with True label and imagery

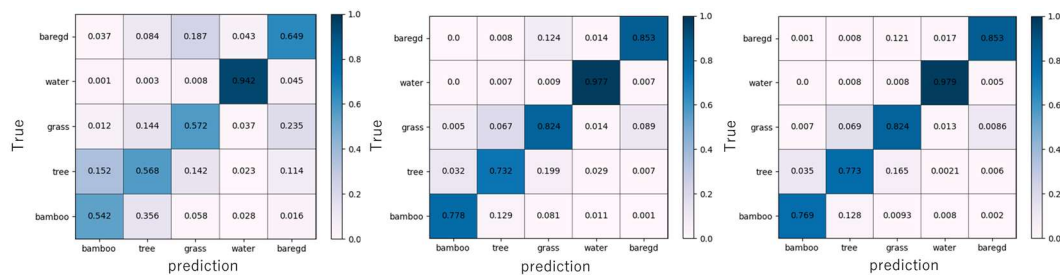


Fig. 3. Confusion matrix of traditional method, Deeplabv3+ model and modified method result

Conclusions

Compared with the traditional method, our modified method showed the Obvious advantages, on the other hand, the recall value of our methodology for the arborous species have been improved approximately 4% than Deeplabv3+ model.

Acknowledgement

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References

Chen, L. *et al.*: Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation, The European Conference on Computer Vision, pp.801-818, 2018.